

Which Arab? How LLMs Treat Candidates from the 22 Arab League Countries in Hiring and Promotion

Project proposal, IS 617 (HWS26)

Laith Sandouk and Aleksander Kasak and Nico Fedotov

University of Mannheim

{laith.sandouk, aleksander.kasak, nico.fedotov}@students.uni-mannheim.de

1 Research questions

Language models have been shown to carry stereotypes about Arabs and Muslims (Section 2). The studies that show this treat them as one group. The 22 Arab League countries, however, differ widely in wealth, in political stability and in how their citizens are seen abroad: high-income Gulf states, middle-income countries such as Egypt, Jordan and Morocco, and countries marked by war and displacement such as Syria, Yemen, Sudan, Libya and Somalia. We ask whether a model sees some of these nationalities more favourably than others. To put a number on this we use hiring and promotion: the model receives a job and CVs that are identical except for one line stating the nationality, and returns a score and a decision.

- RQ1** Do fit scores and decisions differ across the 22 Arab League nationalities when everything else is identical, and is this spread larger than for unrelated placebo nationalities? This is the main question.
- RQ2** Do the differences follow a country's economic and political position, for example Gulf nationalities above nationalities associated with war, displacement or low income?
- RQ3** Does the place of the job matter? Is the local nationality favoured in its own country, and is the same foreign nationality, for example Yemeni or Egyptian, treated differently in a Gulf country than in a North African one?
- RQ4** Do the differences remain when the model is told that nationality is not a criterion, and when it may let a lottery decide between two identical candidates?
- RQ5** Are the patterns the same in hiring and in promotion, and across models?

The comparison between Arab and non-Arab nationalities is secondary.

2 Motivation and related work

Models carry stereotypes about Arabs and Muslims. In GPT-3, 66 of 100 completions of the prompt “Two Muslims walked into a” were violent, significantly more than for other religious groups (Abid et al., 2021), and a weaker form of this association was still found in later GPT models (Hemmatian et al., 2023). In stories written by five models, characters with Middle Eastern or North African names appeared 78% less often in workplace stories than their share of the US population would suggest (Shieh et al., 2026). The two biases are linked. The Arab League countries are majority-Muslim, and when models describe Middle Eastern people they disproportionately mention religion (Cheng et al., 2023). A bias against Muslims can therefore be expected to reach Arab nationalities, and this general disadvantage is our starting point. It has two limits. The studies treat Arabs and Muslims as one group, and they tested models released between 2020 and 2023. Whether the disadvantage appears in the models we test is open; the German, Turkish and no-nationality conditions will show whether it does. Our CVs state a nationality and no religion, so we measure how nationalities are treated, not bias against Islam itself.

Models also tell Arab countries apart. In text generated by GPT-2, Libyan, Sudanese and Tunisian were among the five demonyms with the most negative sentiment (Narayanan Venkit et al., 2023). In simulated immigration decisions, GPT models suggested additional scrutiny for applicants from Iraq, Sudan and Somalia (Mao and Zhao, 2025). Models also represent the values of some Arab countries much better than those of others (Zahraei et al., 2025). None of these studies compares the 22 nationalities with each other in a decision about a person.

Hiring and promotion are our measuring instrument. To compare 22 nationalities we need

a task in which the nationality can change while everything else stays the same, and in which the model answers with a number. A CV does this: the text stays word for word the same, one line changes, and the model returns a fit score and a decision. It is also how discrimination is usually measured, among employers (Lippens et al., 2023) and in LLMs (Gaebler et al., 2024; Armstrong et al., 2024).

The scenario is realistic. Many Arabs work in an Arab country other than their own. In the Gulf states, Arabs from other countries were 72% of the foreign population in 1975 and still about a third in the early 2000s (Kapiszewski, 2006). Nationality is used openly there. A report for the International Labour Organization finds that segmentation in Jordan is based on nationality rather than on education or skills (Razzaz, 2017). Among low-income workers in Qatar, pay differs by nationality (Gardner et al., 2013). Politics has singled out individual Arab nationalities: in 1990–91 more than 1.5 million people were expelled from Gulf states, among them up to one million Yemenis from Saudi Arabia, 200,000 Jordanians and 150,000 Palestinians (Kapiszewski, 2006). An Egyptian accountant applying in Kuwait is an everyday case, and a nationality line on a CV is normal there.

The answer is open in both directions. Field experiments in Western labour markets find about 41% fewer positive callbacks for applicants with Arab, Maghrebi or Middle Eastern names (Lippens et al., 2023), and origin can matter within that group: Swedish employers called back 10% of Iraq-born and 5% of Somalia-born applicants (Vernby and Dancygier, 2019). Inside the region the evidence points the other way. Qatari citizens made no significant distinction between nationalities in a survey experiment on residency (Shockley and Gengler, 2024), and Arab expatriates in Kuwait and Qatar favour “fellow Arabs” as one group (Blaydes and Gengler, 2023). A result of “no difference” would therefore also be an answer.

Existing LLM hiring audits cannot tell. They mostly vary race and gender through names (Gaebler et al., 2024; Armstrong et al., 2024). Where Arab applicants appear, they are one group: Arab names received a penalty of 1.41 points on a 100-point scale in GPT-3.5 (Lippens, 2024). A pooled figure does not say whether the nationalities behind it are treated alike: differences between them are averaged away. The pooling follows from the method, because a name does not identify a coun-

try: raters assigned only 34% of Moroccan names to Morocco (Martiniello and Verhaeghe, 2022). To our knowledge, no study lets an LLM make hiring decisions while varying individual Arab nationalities with qualifications held fixed. The closest one we found pools them into Arab versus non-Arab (Albaroudi et al., 2026). We treat this claim as tentative.

3 Design

Materials. Everything is synthetic; the project audits model behaviour and is not a hiring tool. We prepared 20 job advertisements, from warehouse associate to software developer, and one CV per job. Each CV meets the requirements of its job without being outstanding, so that scores can move in both directions. A CV carries a reference number instead of a name and contains no names of places, schools or companies.

Manipulation. One line in the CV, for example Nationality: Jordanian. There are 29 conditions: the 22 Arab League nationalities, German and Turkish as benchmarks, four unrelated nationalities as a placebo set (Uruguayan, Malawian, Maldivian, Cambodian), and a CV without the line. The placebo set shows how much the scores vary for arbitrary nationality labels.

Setting. The model is told that it recruits for the company in a given country. Each job is placed once in each of the 22 Arab League countries, so every Arab nationality appears both as the local applicant and as a foreign one (RQ3).

Tasks. (1) Rate one applicant: a fit score from 0 to 100 and interview yes or no. (2) Choose between two applicants who have the same CV and different nationalities, with one interview slot. In one version the model may answer A, B or “lottery”; in the other it must choose A or B. Each pair is shown in both orders. (3) Both tasks again with one added paragraph stating that nationality is not a selection criterion. All of this is run in two scenarios: hiring, and an internal promotion with the same job description and the same CV.

Models. Four open-weight models served with vLLM on bwUniCluster: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B and Mistral Small 3.2 24B.

Size. Table 1 lists the runs per model. For the choice task, the 22 Arab nationalities are randomly paired within each job and country. We run the blocks in the order shown and will report which blocks were completed.

Block	Runs per model
1 Hiring, rate one applicant	12,760
2 Hiring, choose between two	19,360
3 Blocks 1 and 2 with the added paragraph	32,120
4 Promotion, all of the above	64,240
Total	128,480

Table 1: Planned runs per model, in order of priority.

4 What we will compare

- RQ1** How far the mean score and the decision rates differ across the 22 Arab nationalities, against the spread across the placebo nationalities.
- RQ2** Each nationality’s result against its country’s income group (World Bank) and against whether the country is affected by armed conflict. The conflict classification still has to be fixed.
- RQ3** Local versus foreign applicants, and each foreign nationality across the countries of the job.
- RQ4** The same comparisons with the added paragraph; in the choice task, which nationality is chosen, how often the model takes the lottery, and what it does when the lottery is removed.
- RQ5** Hiring against promotion, and the four models against each other.

We have not fixed the statistical analysis yet. We will write down the hypotheses and the analysis before the first real run.

5 Timeline

Pitch on 13 October. A timing test and block 1 on all four models before the midterm presentation (11–12 November). The remaining blocks before the final presentation (9–10 December). Report due on 8 January 2027.

6 Questions

1. Is the scope right for a short paper, or should we drop the promotion scenario and keep hiring only?
2. For RQ2, which classification of the countries would you use (income, conflict, region), and is it defensible to group them at all?
3. Is one CV per job at one qualification level enough, or would you expect several levels?

4. Is the choice between two identical CVs with a lottery option a sensible measure, and are four placebo nationalities enough as a yardstick?
5. Are four open-weight models between 8B and 24B parameters acceptable?
6. Do you see concerns with reporting results by individual nationality?

AI use. The job advertisements, CVs and prompts, and a first draft of this proposal, were prepared with the help of Claude Code.

References

- Abubakar Abid, Maheen Farooqi, and James Zou. 2021. [Persistent anti-muslim bias in large language models](#). In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pages 298–306.
- Elham Albaroudi, Taha Mansouri, Mohammad Hatamleh, and Ali Alameer. 2026. [Addressing intersectional bias in AI recruitment using HITHIRE model: a fair, ethical, green AI and transparent hiring solution for Saudi Arabia’s diverse workforce in line with vision 2030](#). *AI and Ethics*, 6(1):57.
- Lena Armstrong, Abbey Liu, Stephen MacNeil, and Danaë Metaxa. 2024. [The silicon ceiling: Auditing GPT’s race and gender biases in hiring](#). In *Proceedings of the 4th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO ’24)*, pages 1–18.
- Lisa Blaydes and Justin Gengler. 2023. [Arab identity and attitudes toward migration in Kuwait and Qatar](#). *Research & Politics*, 10(4).
- Myra Cheng, Esin Durmus, and Dan Jurafsky. 2023. [Marked personas: Using natural language prompts to measure stereotypes in language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1504–1532.
- Johann D. Gaebler, Sharad Goel, Aziz Z. Huq, and Prasanna B. Tambe. 2024. [Auditing large language models for race & gender disparities: Implications for artificial intelligence-based hiring](#). *Behavioral Science & Policy*, 10(2):46–55.
- Andrew Gardner, Silvia Pessoa, Abdoulaye Diop, Kaltham Al-Ghanim, Kien Le Trung, and Laura Harkness. 2013. [A portrait of low-income migrants in contemporary Qatar](#). *Journal of Arabian Studies*, 3(1):1–17.
- Babak Hemmatian, Razan Baltaji, and Lav R. Varshney. 2023. [Muslim-violence bias persists in debiased GPT models](#). *Preprint*, arXiv:2310.18368.
- Andrzej Kapiszewski. 2006. [Arab versus Asian migrant workers in the GCC countries](#). Technical Report

UN/POP/EGM/2006/02, United Nations Population Division, Department of Economic and Social Affairs, Beirut.

Louis Lippens. 2024. [Computer says ‘no’: Exploring systemic bias in ChatGPT using an audit approach](#). *Computers in Human Behavior: Artificial Humans*, 2(1):100054.

Louis Lippens, Siel Vermeiren, and Stijn Baert. 2023. [The state of hiring discrimination: A meta-analysis of \(almost\) all recent correspondence experiments](#). *European Economic Review*, 151:104315.

Yicheng Mao and Yang Zhao. 2025. [Digital gatekeepers: Exploring large language model’s role in immigration decisions](#). *Preprint*, arXiv:2506.21574.

Billie Martiniello and Pieter-Paul Verhaeghe. 2022. [Signaling ethnic-national origin through names? the perception of names from an intersectional perspective](#). *PLoS ONE*, 17(8):e0270990.

Pranav Narayanan Venkit, Sanjana Gautam, Ruchi Panchanadikar, Ting-Hao Huang, and Shomir Wilson. 2023. [Nationality bias in text generation](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 116–122.

Susan Razzaz. 2017. [A challenging market becomes more challenging: Jordanian workers, migrant workers and refugees in the Jordanian labour market](#). Technical report, International Labour Organization, Regional Office for Arab States, Beirut.

Evan Shieh, Faye-Marie Vassel, Cassidy R. Sugimoto, and Thema Monroe-White. 2026. [Intersectional biases in narratives produced by open-ended prompting of generative language models](#). *Nature Communications*, 17(1):1243.

Bethany Shockley and Justin J. Gengler. 2024. [Sharing citizenship: Economic competition, cultural threat, and immigration preferences in the rentier state](#). *Political Science Research and Methods*, 12(1):59–75.

Kåre Vernby and Rafaela Dancygier. 2019. [Can immigrants counteract employer discrimination? a factorial field experiment reveals the immutability of ethnic hierarchies](#). *PLOS ONE*, 14(7):e0218044.

Pardis Sadat Zahraei, Gokhan Tur, Dilek Hakkani-Tür, and Ehsaneddin Asgari. 2025. [The alignment veto: How safety training suppresses cultural knowledge in LLMs](#). *Preprint*, arXiv:2510.13154.